1.11.8.2 Semi-Variable capacitor

Trimmer is a semi-variable capacitor as shown in Figure 1.66.

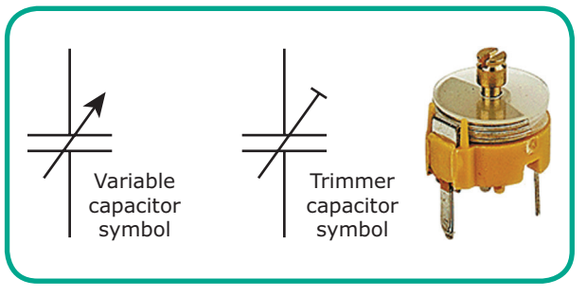


Figure 1.66 Trimmer

In this, a plate is kept static (unmoved) and another plate is placed at a distance and it is in such a way that the second plate can be moved little by little towards the first plate. The second plate is attached with the screw for comfortable movement.

When the plates are nearer, the capacitance is high and when it is at a distance the capacitance is low. This type of capacitor is said to be Trimmer. It is also called as button trimmer. It is available in values from 4pF to 70pF.

Uses

It is used in pre-tuning circuits of the radio receivers.

1.11.9 Capacitor in series

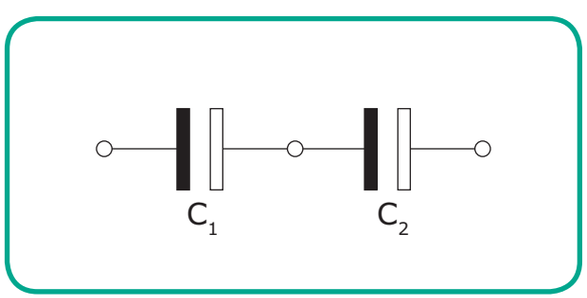


Figure 1.67 Capacitors in series

Figure 1.67 shows the capacitors connected in series.

When the capacitors are connected in series as shown in Figure 1.67, the total capacitance is

$$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2}$$

$$C = \frac{C_1 C_2}{C_1 + C_2}$$

Problems

Find out the total capacitance, when two capacitors of values 10 μF, 15 μF are connected in series.

Total capacitance

$$C = \frac{C_1 C_2}{C_1 + C_2}$$

$$C = \frac{10 \times 15}{25}$$

Total capacitance C = 6 μF

1.11.10 Capacitors in Parallel

Figure 1.68 shows a circuit in which the capacitors are connected in parallel.

When the capacitors are connected in parallel as shown in Figure 1.68, the total capacitance is $C = C_1 + C_2$

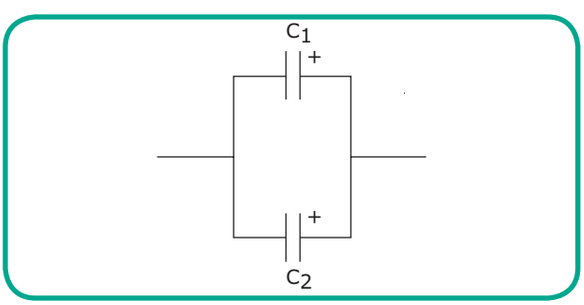


Figure 1.68 Capacitors in parallel

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